



UNIVERSITY OF DHAKA

Institute of Leather Engineering and Technology
Department of Leather Engineering

LABORATORY REPORT

Course Code: **LE-410**

Course Title: Wastewater and Soil Waste Management Lab

B.Sc. in Leather Engineering· 4th Year 2nd Semester· Examination 2025

EXPERIMENT NO. 01

Characterization of Physicochemical Parameters of Tanning Effluent (Lime Liquor)

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Group number	A
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Submitted to

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DECLARATION OF ORIGINALITY

I declare that this laboratory report is my own work.

I understand and accept the following:

- The experimental readings were obtained by my group, and the group's raw data and the pooled class data may legitimately be identical to those of my group members and classmates.
- Everything else in this report - the theory, the procedure as I have written it, the calculations, the discussion and the conclusion - must be written in my own words and represent my own understanding.
- Copying text from another student, from a previous year's report, or from any source without acknowledgement is plagiarism.
- All reports submitted for this course are compared with one another. Where the written sections of two or more reports are found to be substantially the same, all the students concerned will be penalised, whatever the source of the copying.

Scanned Signature of the student:  Date: 08-09-2026

How this report will be assessed for originality

Data may be shared. Understanding may not. Two students of the same group are expected to submit the same numbers and different reports.

The sections most often copied - and most carefully checked - are the Theory, the Discussion and the Conclusion. Write these yourself, in your own sentences, however imperfect. A short, honest paragraph in your own words is worth far more than a fluent one that is not yours.

1. OBJECTIVE

The objective of this experiment was to determine the physicochemical characteristics of tannery effluent by measuring its pH, total dissolved solids (TDS), and alkalinity. These are the primary pollution parameters were used to assess the chemical nature and pollution load of the lime liquor generated during liming operation.

2. THEORY AND PRINCIPLE

Lime liquor is a wastewater stream generated mainly during the liming operation of leather processing. During this operation, the hides/skins are treated with an alkaline mixture, commonly contain calcium hydroxide, sodium sulfide, to remove hair, open up the fibrous structure. As a result, the spent lime liquor generally contains organic matters, suspended materials, and alkaline compounds. The physicochemical characterization of the effluent is important for understanding its chemical nature and assessing its potential effect on wastewater treatment.

pH is a measure of the hydrogen-ion activity of an aqueous solution and indicates whether the solution is acidic, neutral, or alkaline. Total dissolved solids (TDS) represent the total concentration of dissolved substances present in water or wastewater. These substances may include soluble salts, inorganic ions, and small amount of dissolved organic matter. The pH and TDS were determined using multiparameter device. Alkalinity is the capacity of water to neutralize added acid. It is mainly associated with hydroxide (OH^-). Since lime liquor contains alkaline chemicals, its alkalinity can be considerably higher than of ordinary water.

The alkalinity of the sample can be determined by acid titration. A known volume of lime liquor should be titrated against a standard acid solution until the required endpoint obtain. The amount of acid consumed is related to the alkalinity of the sample.

The general titration relationship is:

$$\text{Alkalinity (as CaCO}_3\text{, mg/L)} = \frac{V \times N \times 50,000}{V_s}$$

Where:

V = volume of acid used in titration (mL)

N = normality of the acid

V_s = volume of sample taken for titration (mL)

50,000 = equivalent weight factor for CaCO_3 expressed in mg/eq.

3. APPARATUS AND REAGENTS

3.1 Apparatus and instruments

1. Multiparameter
2. Beaker of 500mL
3. Glass Rod
4. Burette

3.2 Reagents and materials

1. Waste-water
2. Standard Sulfuric acid
3. Phenolphthalein Indicator
4. Methyl Orange Indicator

4. PROCEDURE

4.1 Multiparameter test

1. 300mL of waste-water was taken into a clean 500mL beaker.
2. The probe of the multiparameter was immersed in the beaker containing the waste-water
3. It was left for a while to stabilize meter reading. The pH and TDS value were recorded.

4.2 Titration for alkalinity test

1. 50mL filtered lime liquor sample was taken into conical flask
2. 2-3 drops of phenolphthalein indicator was added into the sample.
3. Using sulfuric acid the sample was titrated until the pink color completely disappear.
4. Then 2-3 drops of methyl orange indicator was added.
5. The titration was continued until the color changes sharply from orange to red. The total used acid volume was recorded

5. OBSERVATIONS AND DATA

5.1 Sample and conditions

Sample / stream tested	Lime liquor
Variable assigned to your group	
Date and time of sampling	02-08-2026
Temperature	28°C

5.2 Your group's data

Table 1: Reading from Multiparameter

Parameter / Trial	Reading 1
pH	9.9
Total Dissolved Solids (TDS)	3284ppm
Electrical Conductivity (EC)	6428 μ S/cm
Sample Volume	300ml

Table 2: Reading for alkalinity test

Parameter / Trial	Reading 1
Sample Volume (V _s)	50mL
Sulfuric Acid Conc.(N)	1N
Volume of Sulfuric Acid required(V)	2.1mL

6. CALCULATIONS

6.1 Formula used

$$\text{Alkalinity (as CaCO}_3\text{, mg/L)} = \frac{V \times N \times 50,000}{V_s}$$

Where:

V = volume of acid used in titration (mL)

N = normality of the acid

V_s = volume of sample taken for titration (mL)

50,000 = equivalent weight factor for CaCO₃ expressed in mg/eq.

6.2 Specimen calculation

$$\begin{aligned} \text{Alkalinity} &= (2.1 \times 1 \times 50000)/50 \\ &= 2100 \text{ mg/L as CaCO}_3 \end{aligned}$$

6.3 Summary of calculated results

Quantity	Value	Unit	Remarks
pH	9.9	pH Scale	Strong alkaline
TDS	3284	ppm	Higher TDS
EC	6428	μS/cm	Higer ionic mobility
Total Alkalinity	2100	mg/L as CaCO ₃	Strong alkaline

7. GRAPHS

8. RESULTS

Parameter determined	Result obtained	Standard (WHO)
pH	9.9	6.5-9.0
TDS	3284ppm	<600ppm
EC	6428	
Alkalinity	2100 mg/L	

9. DISCUSSION

The physicochemical characterization of the lime liquor showed that the effluent was strongly alkaline and contained a considerable amount of dissolved substances. The measured pH was 9.9, which indicated an alkaline condition. This alkaline nature was expected because alkaline chemicals such as lime and sodium sulfide were used during the liming operation. The measured TDS was 3284 ppm, indicating a high concentration of dissolved substances in the effluent. These dissolved materials could have originated from the chemicals used during processing as well as substances extracted from the hides or skins.

The total alkalinity of the lime liquor was found to be 2100 mg/L as CaCO_3 . This relatively high alkalinity indicated that the sample had a considerable capacity to neutralize acid. The result was consistent with the alkaline nature of lime liquor and supported the pH measurement. The combination of high pH and high alkalinity showed that the effluent would require appropriate treatment before discharge.

According to the standard values recorded in the report, the measured pH of 9.9 was higher than the stated range of 6.5–9.0, while the TDS value of 3284 ppm was considerably higher than the stated limit of 600 ppm. Therefore, the sample did not satisfy these indicated limits. The high dissolved-solids concentration and alkaline condition suggested that direct discharge of the untreated lime liquor could adversely affect the receiving environment and therefore required suitable wastewater treatment.

Some variation in the measured values could have resulted from inadequate cleaning of the multiparameter probe, insufficient stabilization of the instrument reading, or non-uniform mixing of the effluent before sampling. During the alkalinity determination, an endpoint error could also have affected the calculated value. Adding excess sulfuric acid beyond the endpoint would increase the recorded acid volume and consequently produce a higher calculated alkalinity, whereas stopping before the true endpoint would produce a lower value.

10. CONCLUSION

The physicochemical characteristics of the lime liquor were determined through measurements of pH, TDS, and alkalinity. The sample showed a pH of 9.9, TDS of 3284 ppm, and total alkalinity of 2100 mg/L as CaCO₃. These results indicated that the lime liquor was strongly alkaline and contained a high concentration of dissolved substances. Therefore, the effluent required appropriate treatment before discharge.

11. PRECAUTIONS AND SAFETY

1. The lime liquor sample was handled carefully to avoid direct contact with the skin and eyes.
2. Laboratory gloves and a lab coat were worn during sample handling and titration.
3. The multiparameter probe was cleaned and rinsed properly before and after measurement to prevent contamination and cross-contamination.
4. The probe was immersed properly in the sample without allowing it to touch the bottom or sides of the beaker.
5. Sufficient time was allowed for the multiparameter reading to stabilize before recording the pH and TDS values.
6. The sample was mixed properly before measurement so that the collected portion represented the effluent adequately.
7. Sulfuric acid was handled carefully because it is corrosive and can cause burns on contact with skin or eyes.
8. The sulfuric acid was added carefully from the burette to avoid splashing and overshooting the titration endpoint.
9. The phenolphthalein and methyl orange indicators were handled carefully and unnecessary contact with them was avoided.
10. The burette was checked for leakage and air bubbles before starting the titration.
11. The titration endpoint was approached slowly to reduce the possibility of adding excess acid.
12. Any spilled sample or reagent was cleaned immediately according to laboratory safety procedures.
13. The tested wastewater and chemical residues were disposed of through the designated laboratory wastewater disposal system and were not released directly into the environment.

12. REFERENCES

1. World Health Organization (WHO). Guidelines for Drinking-water Quality: Fourth Edition Incorporating the First and Second Addenda. Geneva: World Health Organization, 2022. ISBN 978-92-4-004506-4.